torch.blackman_window#

https://pytorch.org/docs/stable/generated/torch.blackman_window.html

Description:

Blackman window function. where NNN is the full window size. The input window_length is a positive integer controlling the returned window size. periodic flag determines whether the returned window trims off the last duplicate value from the symmetric window and is ready to be used as a periodic window with functions like torch.stft(). Therefore, if periodic is true, the NNN in above formula is in fact window_length+1\text{window_length} + 1window_length+1. Also, we always have torch.blackman_window(L, periodic=True) equal to torch.blackman_window(L + 1, periodic=False)[:-1]. If window_length =1=1=1, the returned window contains a single value 1.

Function Signature

```
torch.blackman_window(window_length, periodic=True, *,
dtype=None, layout=torch.strided, device=None,
requires_grad=False) → Tensor#

Parameters

Keyword Arguments

Returns

Return type
```

Parameters

Keyword

dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). Only floating point types are supported. layout (torch.layout, optional) – the desired layout of returned window tensor. Only torch.strided (dense layout) is supported. device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.

Returns:

Tensor

Notes & Warnings

NOTE Note If window_length =1=1=1, the returned window contains a single value 1.

Detailed Documentation

Documentation

Blackman window function.

where NNN is the full window size.

The input window_length is a positive integer controlling the returned window size. periodic flag determines whether the returned window trims off the last duplicate value from the symmetric window and is ready to be used as a periodic window with functions like torch.stft(). Therefore, if periodic is true, the NNN in above formula is in fact window_length+1\text{window_length} + 1window_length+1. Also, we always have torch.blackman_window(L, periodic=True) equal to torch.blackman_window(L + 1, periodic=False)[:-1].

If window_length =1=1=1, the returned window contains a single value 1.

window_length (int) – the size of returned window

periodic (bool, optional) – If True, returns a window to be used as periodic function. If False, return a symmetric window.

dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). Only floating point types are supported.

layout (torch.layout, optional) – the desired layout of returned window tensor. Only torch.strided (dense layout) is supported.

device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.

A 1-D tensor of size (window_length,)(\text{window_length},)(window_length,) containing the window

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